COMPUTER ENGINEERING

QUEZON CITY

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Assignment 6.1 **Linear and BInary Search**

1. PROCEDURE OUTPUT

1. What is a search tree in data structures?

A node-based hierarchical data structure with nodes each containing a key in a strict ordering invariant is called a search tree. A typical Binary Search Tree (BST) will have the key value of its left subtree less than the key value of the parent node, and the right subtree more. This structural rule permits search algorithms to be able to eliminate subtrees at every step that are irrelevant to the search, which therefore greatly reduces search space.

2. What are the different types of search algorithms in data structures? Differentiate each type of search.

Algorithm	Required Data Structure	Average Time Complexity	Core Mechanism
Linear Search	Unsorted or Sorted Array/Lists	$O(n)$	Scan the elements one by one from start till finding the target element.
Binary Search	Sorted, Random Access arrays	$O(log\ n)$	Works to narrow down the search space by comparing the target with the middle element, repeatedly until the desired element is found.

Tree Search	Search Trees (BST, AVL, B-Trees)	$O(log\ n)$	Resets to the rear end of the tree (along certain pointer paths, depending on the keys of nodes).
Harsh Search	Harsh Tables	$O(1)$	Calculates a direct memory index into a mathematical hash function.
Graph Search	Graph/ Networks	$O(V\ +\ E)$	Traverses level-wise of the nodes (Breadth First) or traverses branch-wise of the nodes (Depth First).

3. What operations / implementations can be performed using binary and linear search operations?

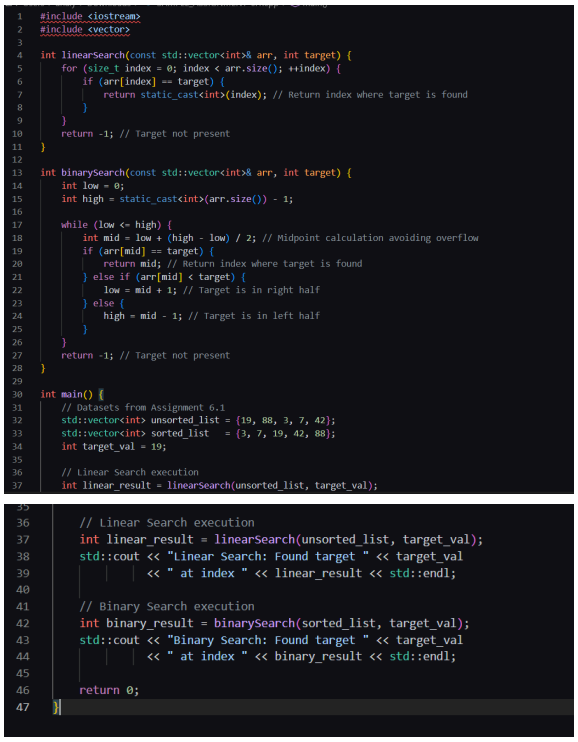
Linear search is usually applied to search unsorted data, and traverse linked lists. It is also very powerful for finding global minimums or maximums and for small arrays performing "single-pass" operations. However, binary search is optimized for searching in sorted arrays and finding the lower/upper bound, as well as floor/celling values. In addition, binary search can be used in range queries and be used to solve numerical search space optimization problems, like `git bisect`.

4. What are the advantages in using binary search tree as data structure?

Dynamic search trees maintain an average $O(log\ n)$time complexity for fast searching, insertion, and deletion operations. By using dynamic pointers, they grow and shrink on demand to completely avoid the memory overhead of array reallocation. An in-order traversal retrieves all stored elements in sorted order in $O(n)$ time without requiring an additional sorting pass. These structures also provide flexible navigation, easily pinpointing minimums, maximums, immediate predecessors, and successors. Finally, this ordered layout enables efficient bounded queries to retrieve all elements residing

within a specified range.

5. Give an example program using binary search and Linear search.

INPUT	OUTPUT
	

```
C:\msys64\ucrt64\bin>C:\Users\analy\Downloads\SAMPLE_ASSIGNMENT-6.1.exe
Linear Search: Found target 19 at index 0
Binary Search: Found target 19 at index 2
```

CONCLUSION:

Search algorithms and data structures are crucial for accessing and managing data effectively, depending on the nature of the data. Linear search is an algorithm that searches for an element linearly through an array or linked list, with a time complexity of $O(n)$. Binary search

uses a sorted, random-access array to improve search time to $O\ (log\ n)$. These ideas are the basis of more complex data structures such as binary search trees, which also use dynamic memory allocation and enjoy logarithmic time complexity for insertion, deletion, and range queries. The above C++ program was successfully coded for demonstrating these concepts with unsorted and sorted data sets and used linear search to find the target element in the unsorted data set and binary search in the sorted data set. Finally, the choice of optimal search method depends on the type of data organization, memory overhead, and runtime efficiency needed for the specific application at hand.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Quillbot**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Fix my grammar and paraphrasing**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I only used it to fix my grammar and paraphrase some of my words but all of my answers are all mine and knowledge**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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